

Unit -1& 2

2 Marks

1. Define the role of statistics in data science.
2. What are numerical and categorical variables? Give one example of each.
3. Define a random variable.
4. What is the difference between discrete and continuous data?
5. What is a sample in statistics?
6. Name two types of sampling methods.
7. Define the term "mean" in statistics.
8. How is the median of a dataset calculated?
9. What is the mode in a dataset?
10. Define range as a measure of dispersion.
11. What is standard deviation?
12. What is variance, and how is it related to standard deviation?
13. Define skewness in data distribution.
14. What is kurtosis?
15. What is covariance?
16. Define correlation and explain its range of values.
17. Define permutation and combination.
18. What is the fundamental difference between permutation and combination?
19. List the different types of probability.
20. What is classical probability?
21. Define marginal probability.
22. What is conditional probability?
23. State Bayes Theorem.
24. Give one application of Bayes Theorem.
25. What is a binomial distribution?
26. What is a Poisson distribution used for?
27. Define normal distribution.
28. What is the total area under a normal distribution curve?
29. State the multiplication rule for probability.
30. How does independent probability differ from dependent probability?
31. What is the expected value in a probability distribution?
32. Give an example of real-life usage of the Poisson distribution.
- 33.

16 Marks

1. Explain the role of statistics in data science and discuss how it aids in data-driven decision-making with examples.
2. Discuss in detail the different types of data: numerical, categorical, and random variables, and how they are utilized in data analysis.
3. What are the key methods of data collection? Explain the different types of sampling with examples.
4. Define and explain the measures of central tendency (mean, median, and mode) and their importance in data analysis with relevant examples.
5. Discuss the measures of dispersion: range, quartiles, standard deviation, and variance. How do they help in understanding the spread of data?
6. Explain the concepts of skewness and kurtosis in the distribution of data. Provide examples of different types of skewness and kurtosis.
7. Define covariance and correlation. Discuss their differences and explain their significance in understanding relationships between variables with examples.
8. How do covariance and correlation help in determining the strength and direction of the relationship between two variables? Illustrate with real-world examples.
9. Explain permutation and combination in detail with examples. Highlight the differences between them and provide real-life applications.
10. Discuss the different types of probability: classical, empirical, and subjective. Provide examples for each.
11. Explain the rules for computing probability, including the addition and multiplication rules, with appropriate examples.
12. Define marginal and conditional probability. Discuss their significance with the help of real-world examples.
13. Discuss Bayes Theorem in detail. Explain its formula and derive it with the help of a practical example.
14. Explain how Bayes Theorem is applied in real-world scenarios. Provide examples to demonstrate its usage in fields such as medical diagnosis or spam filtering.
15. Discuss the binomial distribution, its properties, and its application in solving real-world problems. Include relevant examples and mathematical explanations.
16. Compare and contrast the binomial, Poisson, and normal distributions. Provide examples where each distribution would be most appropriately used.
17. Explain the normal distribution in detail. Discuss its properties, significance in statistics, and applications in real-life data analysis.

Note : Try to learn wherever formulas are applicable and also practice problems related to it.

